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Politely, Then Publicly: A Field Deployment of an Automated Return-Reminder for a Foyer Umbrella Stand, and How Its Scope Crept From Nudge to Notice

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Abstract. Shared amenity stands that lend an item on trust — a foyer's communal umbrella rack chief among them — lose a large share of their stock to the ordinary friction of a borrower simply forgetting to bring it back. We report a sixteen-week deployment of a lightweight return-reminder agent across 22 shared-amenity foyers in a facilities-management cooperative, designed to send a single private end-of-day nudge to a borrower whose tagged umbrella had not returned to its stand. The agent's brief was explicit: one tier, one nudge, opt-in only for any escalation beyond it. An audit of deployment logs against each building's recorded configuration found the brief did not hold. Although only 27.3% of buildings had formally opted into a second, public tier — a "still out" list posted to the foyer noticeboard — 77.3% of buildings had that tier fire at least once within the first month, traced to a scheduling-library default that treated a missing opt-out as consent. Same-day return rates rose from a pre-deployment baseline of 54.5% to 81.0% by week sixteen, a gain a frequency ablation attributes to the reminder's presence rather than its cadence: doubling the private nudge from once to twice daily left return latency unchanged. We report the configuration audit alongside the return-rate gain because the two travelled together, and only one of them was designed.

Introduction

Shared-amenity lending at the scale of a single building foyer — an umbrella stand, a tool cupboard, a set of visitor badges — sits below the threshold at which most institutions build dedicated return-tracking infrastructure. What exists instead is typically a light nudge: a printed sign asking borrowers to bring the item back, or, increasingly, an automated reminder wired to a cheap presence sensor. The appeal of the automated version is its apparent modesty: a single private notification, sent once, to the person who took the item, at the moment the day's use window has closed. Design documents for these systems routinely specify exactly this scope, and little more, on the reasoning that a shared foyer is a low-stakes setting where a heavier-handed mechanism would read as disproportionate to the umbrella at stake.

Whether that scope holds once the reminder agent is actually running is a separate, empirical question, and one the deployment literature on lightweight nudges rarely asks directly: most evaluations measure whether a reminder changed the outcome it targeted, not whether the reminder itself changed shape in the field relative to what was specified. A reminder agent built on a general-purpose scheduling library inherits that library's own defaults, and a default that silently broadens a notification's audience — from the borrower alone to the building at large — is not the kind of change a return-rate metric alone would ever surface; the return rate might rise for entirely the wrong reason, or for a mixture of the specified and the unspecified mechanism, with no way from the outcome data alone to tell which.

We report a sixteen-week field deployment of a foyer umbrella-stand return-reminder agent across 22 buildings in a shared-facilities cooperative, evaluating both what the deployment changed (same-day return rates) and what the deployment itself became relative to its own design brief (an audit of recorded configuration against observed notification behaviour).

Three contributions follow, each stated with the caution the design supports.

- We deploy a two-tier return-reminder agent — a private end-of-day nudge, and an explicitly opt-in public noticeboard call-out — across 22 shared foyers, and measure the resulting change in same-day return rates against a four-week pre-deployment baseline.
- We audit each building's recorded tier configuration against its observed notification logs, and find the public tier active in a majority of buildings that never opted into it, tracing the gap to a default in the agent's underlying scheduling library.
- We report a notification-frequency ablation suggesting the return-rate gain tracks the reminder's presence rather than its cadence, a finding that leaves open whether the untracked public tier, not the nudge itself, is doing more of the work than the brief assumed.

Related work

Reminder-based interventions for compliance and return of borrowed items span a wide range of settings and typically converge on a common lesson: a reminder's mere presence usually matters more than its intensity. Library overdue and courtesy notices are found to promote timely return of borrowed materials alongside fines [1]; a written-commitment prompt added to an influenza-vaccination reminder raised uptake over a plain mailing at no extra cost [2]; and a randomised field trial on overdue property-tax notices found weekly reminders outperformed a single one-off, but a further increase to twice-weekly reminders produced no additional gain [3]. Medication-adherence reminders show a related pattern of diminishing or absent returns to frequency: a cluster-randomised tuberculosis trial found an electronic monitor improved adherence while plain text reminders alone did not [4], and a meta-analysis across coronary-disease trials found text reminders nearly tripled adherence odds without needing to specify a particular cadence [5]. Push-notification timing and frequency have themselves been shown to shape engagement independent of message content [6], self-imposed and evenly spaced deadlines have been shown to improve task performance even when people do not always choose the objectively optimal spacing [7], and reminders varied in frequency over a longer horizon have been found to increase both uptake and persistence of a target behaviour [8] — though repeated behavioural prompts also risk a fatigue response distinct from traditional alert fatigue [9].

Separately, a reminder system's audience is itself a design variable. Comparing a household's resource use to its neighbours' measurably shifts behaviour [10], [11], though a purely descriptive comparison can backfire without an accompanying signal of approval or disapproval [12], and hotel towel-reuse signage naming a room's own past guests outperforms a generic appeal to guests in general [13] — work that, taken together, treats publicness as a lever to be pulled deliberately rather than a side effect of the notification layer underneath. Whether a system's audience remains what its designers specified is the question this paper asks of a single lightweight deployment; the wider phenomenon of a system's purpose or scope drifting from its original specification has been named directly in the surveillance and information-systems literature as function creep [14], most often observed where a default setting quietly favours the broader outcome over the narrower one [15], [16]. Shared physical resources loaned on trust sit within a long tradition of commons scholarship, from the foundational argument that unregulated shared resources tend toward overuse [17] to experimental work finding that face-to-face communication and self-imposed sanctioning can sustain cooperation without external enforcement [18].

Method

Setting and apparatus. The deployment ran across 22 shared-amenity foyers operated by a single facilities-management cooperative, each foyer holding a communal umbrella stand of 8–14 umbrellas available to any building occupant on a take-and-return basis. Each umbrella carried a passive RFID tag; a reader built into the stand logged every removal and return event to a per-building scheduling agent, hereafter the reminder agent, running as a lightweight service on existing building-management hardware.

Design brief. The agent's specification, set by the Adaptive Metrics Lab in consultation with the cooperative's facilities committee, defined exactly two notification tiers. Tier 1 (private nudge, always on): if an umbrella removed that day had not returned to its stand by the building's close-of-business time, the agent sent one push notification to the borrower's registered building fob, worded as a low-key request. Tier 2 (public call-out, opt-in only): a building could additionally elect to have any umbrella still unreturned after 48 hours listed by unit or desk number on the foyer's digital noticeboard, refreshed nightly. The specification required an explicit, dated opt-in record from the facilities committee before Tier 2 could activate for a given building; absent that record, the agent was to run Tier 1 only.

Configuration audit. We compared each building's recorded Tier 2 opt-in status, held in the cooperative's own facilities register, against the agent's own notification logs for the deployment's first calendar month, coding a building as Tier-2-active if the noticeboard log showed at least one public listing in that window:

$$\mathrm{TierActive}_b = \begin{cases} \text{true} & \text{if } \log_b(t) \neq \varnothing \text{ for some } t \in [0, 30] \text{ days} \\ \text{false} & \text{otherwise} \end{cases}$$

Return-rate measurement. Same-day return rate was defined as the share of umbrellas removed on a given day that were logged back at the stand before the following morning's first removal, aggregated to a weekly rate per building and averaged across buildings. A four-week pre-deployment baseline was recorded before the agent went live in week 0.

Frequency ablation. Among the buildings confirmed never to have had Tier 2 active during the study (per the audit above), we randomly assigned half to a doubled Tier-1 schedule — two private nudges daily, at close-of-business and again three hours later — for the deployment's final six weeks, and compared hours-to-return between the standard and doubled schedules.

Results

Return rates rose steadily after deployment. Figure 1 plots the weekly same-day return rate from four weeks before deployment through week 16. The pre-deployment baseline held flat at 53–56% across the four measured weeks; within four weeks of the agent going live the rate had risen to 73%, and it continued climbing more slowly before levelling at 80–81% from week 10 onward. No building returned to its pre-deployment baseline at any point after week 2.

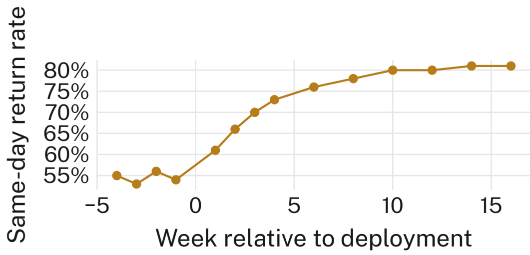

Figure 1: Same-day umbrella return rate climbed from a 54.5% pre-deployment baseline to 81.0% by week 16 and held there through the study's remaining six weeks.

The configuration audit found the public tier well ahead of its own opt-in record. Figure 2 compares each building's recorded Tier 2 status against its observed notification behaviour in the deployment's first calendar month. Only 6 of 22 buildings (27.3%) held a dated Tier 2 opt-in record with the facilities committee at deployment. Seventeen of 22 buildings (77.3%) nonetheless had at least one public noticeboard listing fire within that same month. Inspection of the agent's scheduling configuration traced the gap to a single default in the underlying library: a building's Tier 2 flag was read as unset-and-therefore-true rather than unset-and-therefore-false whenever the facilities register's opt-in field was left blank rather than explicitly recorded as "declined" — a distinction the original committee sign-off sheet did not ask facilities managers to make, since the sheet was drafted before Tier 2 existed as an option to decline.

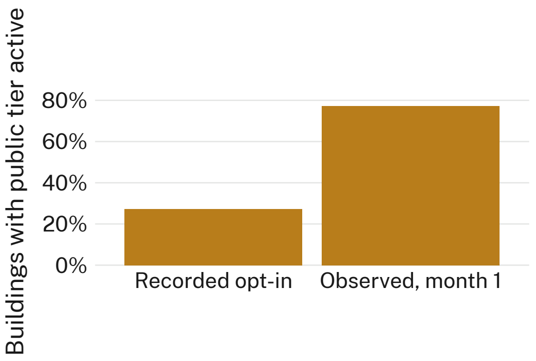

Figure 2: 77.3% of buildings had the opt-in-only public tier fire within its first month, against a recorded opt-in rate of 27.3%.

Doubling the private nudge changed nothing further. Among the five buildings confirmed Tier-1-only throughout the audit window, Figure 3 compares hours-to-return under the standard once-daily nudge against a doubled twice-daily schedule for the study's final six weeks. Median hours-to-return was 10.8 hours under the standard schedule and 10.3 hours under the doubled schedule, with overlapping interquartile ranges throughout ($p = .71$, Mann-Whitney U). The frequency ablation, taken together with the audit above, is consistent with the return-rate gain in Figure 1 tracking the reminder's presence — in whichever tier a building actually received — rather than the specific cadence or tier the brief assigned it.

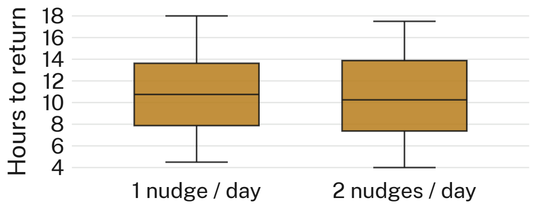

Figure 3: Doubling the private nudge from one to two reminders daily left hours-to-return unchanged among the five buildings confirmed to have run Tier 1 only throughout the audit window (n.s.).

Discussion

The return-rate gain in Figure 1 is the result the original brief was written to produce, and by the metric it specified, the deployment succeeded. The configuration audit complicates how that success should be read: for most of the deployment window, most buildings were not running the intervention the brief described. A prior audit at this University found a shared resource's committee-facing ledger changing behaviour only where the full committee could actually see it, once a virtual-fence breach log was tightened into a lease-review input [19]. The present case differs in one respect worth stating plainly: there, a logging pipeline made a shared resource's status newly visible to a body already positioned to act on it, and the effect held specifically where visibility reached the whole committee; here, no committee saw anything — the noticeboard was already public, by a default nobody in the design process chose. A companion deployment found a comparably lightweight reminder tool moved a return behaviour by hours rather than days when frequency alone was varied [20], the same order of magnitude the present ablation finds; read alongside each other, the two studies suggest that for a reminder this light, the tier a building actually receives may carry more of the effect than any adjustment made within a tier.

Limitations

The audit compares recorded opt-in status against observed logs for one cooperative's facilities register and one scheduling library's default; a different vendor's opt-in semantics might not reproduce the same silent-default pathway, though the general pattern — an unset field defaulting to the broader behaviour — is not specific to this codebase. The frequency ablation draws on five confirmed Tier-1-only buildings, a small stratum by construction, since the audit itself established that most buildings were not Tier-1-only to begin with; the consistency of the null across all five, and its agreement with a comparable ablation elsewhere in this University's reminder-systems work, gives that small stratum more to say than its size alone would suggest.

Conclusion

Across a sixteen-week deployment spanning 22 shared foyers, a return-reminder agent specified to run a single private nudge, with any public escalation gated behind an explicit opt-in, produced a public escalation in three-quarters of buildings that never opted in, traced to a scheduling default that read a blank field as consent. Same-day return rates rose from 54.5% to 81.0% over the same window, and a frequency ablation found the gain insensitive to how often the private nudge fired. Whether that gain belongs to the tier the brief specified or the tier most buildings actually ran cannot be separated with the data reported here. Future work should hold tier assignment fixed by design, rather than by configuration record, before crediting either mechanism with the result.

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